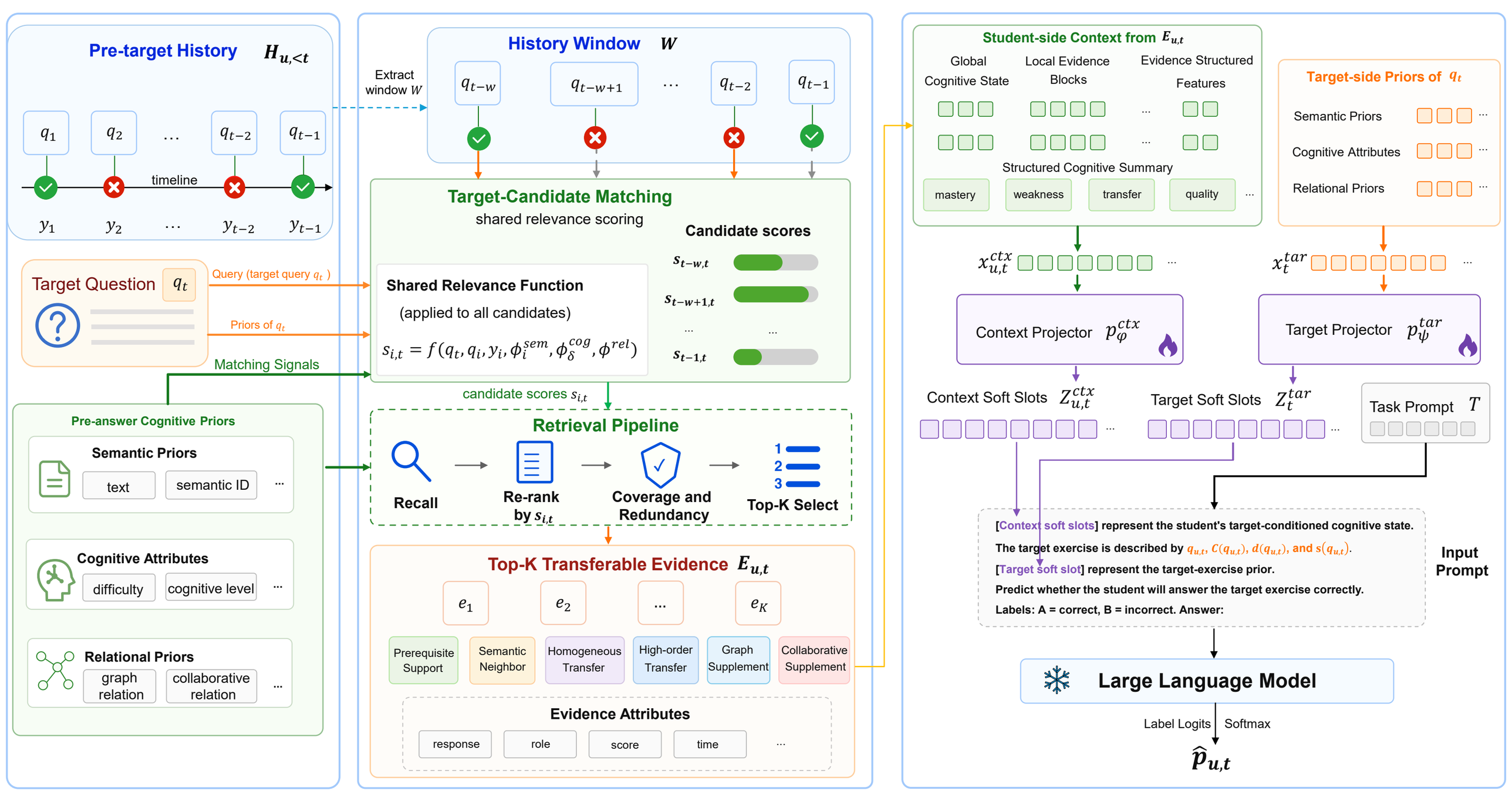

Pre-target History $H_{u,<t}$
q_1 q_2 ... q_{t-2} q_{t-1}
timeline
y_1 y_2 ... y_{t-2} y_{t-1}
Target Question q_t
Query (target query q_t)
Priors of q_t
Matching Signals
Pre-answer Cognitive Priors
Semantic Priors
text
semantic ID
Cognitive Attributes
difficulty
cognitive level
Relational Priors
graph relation
collaborative relation
Extract window W
History Window W
q_{t-w} q_{t-w+1} ... q_{t-2} q_{t-1}
Target-Candidate Matching
shared relevance scoring
Shared Relevance Function
(applied to all candidates)
$s_{i,t} = f(q_t, q_i, y_i, \phi_i^{sem}, \phi_\delta^{cog}, \phi^{rel})$
Candidate scores
$s_{t-w,t}$
$s_{t-w+1,t}$
$s_{t-1,t}$
candidate scores $s_{i,t}$
Retrieval Pipeline
Recall
Re-rank by $s_{i,t}$
Coverage and Redundancy
Top-K Select
Top-K Transferable Evidence $E_{u,t}$
e_1 e_2 ... e_K
Prerequisite Support
Semantic Neighbor
Homogeneous Transfer
High-order Transfer
Graph Supplement
Collaborative Supplement
Evidence Attributes
response
role
score
time
Student-side Context from $E_{u,t}$
Global Cognitive State
Local Evidence Blocks
Evidence Structured Features
Structured Cognitive Summary
mastery
weakness
transfer
quality
Target-side Priors of q_t
Semantic Priors
Cognitive Attributes
Relational Priors
$x_{u,t}^{ctx}$
x_t^{tar}
Context Projector p_φ^{ctx}
Target Projector p_ψ^{tar}
Context Soft Slots $Z_{u,t}^{ctx}$
Target Soft Slots Z_t^{tar}
Task Prompt T
[Context soft slots] represent the student's target-conditioned cognitive state.
The target exercise is described by $q_{u,t}$, $C(q_{u,t})$, $d(q_{u,t})$, and $s(q_{u,t})$.
[Target soft slot] represent the target-exercise prior.
Predict whether the student will answer the target exercise correctly.
Labels: A = correct, B = incorrect. Answer:
Input Prompt
Large Language Model
Label Logits
Softmax
$\hat{p}_{u,t}$
1. Task Input
2. Target-conditioned Evidence Retrieval
3. Context Construction & Frozen LLM Prediction